

# Quantitative Methods in Cell and Molecular Biology

MOL 518

Genomics

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## What is Genomics?

- Sequencing, analysis, evolution, and comparison of ENTIRE genomes

| Organism                                   | Genome Size | Genes  | Year |
|--------------------------------------------|-------------|--------|------|
| <i>Escherichia coli</i> (bacteria)         | 4.6 Mbp     | 4.3 k  | 1997 |
| <i>Saccharomyces cerevisiae</i> (yeast)    | 12.4 Mbp    | 5.7 k  | 1996 |
| <i>Caenorhabditis elegans</i> (worm)       | 100.2 Mbp   | 21.7 k | 1998 |
| <i>Drosophila melanogaster</i> (fruit fly) | 122.6 Mbp   | 17 k   | 2000 |
| <i>Arabidopsis thaliana</i> (plant)        | 135 Mbp     | 27.4 k | 2000 |
| <i>Homo sapiens</i> (humans)               | 3.3 Gbp     | 21 K   | 2001 |

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## Why is it important to study Genomics?

1. To understand basic human biology, development, anatomy, and physiology (tissue-level expression and mutations, genes involved in specific functions, genetic traits)
2. Model organisms – worms, flies, plants, mice
3. To study evolution, Neanderthals and Human, Woolly Mammoth, Dinosaurs
4. To understand diseases (diagnosis and treatment) – GWAS studies, pharmacogenomics, personalized medicine
5. To study pathogens (bacteria, fungi, viruses) and design treatments (resistance)
6. Diagnostic tests: COVID-19, prenatal screening (genetic disorders), Forensics, etc.
7. To engineer model organisms to perform beneficial functions (biofuel production, crop enhancement, drug production, etc.)
8. To enable other disciplines and methodologies: proteomics, CRISPR screens, single cell-sequencing, transcriptomics

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## Why study Computational Genomics?

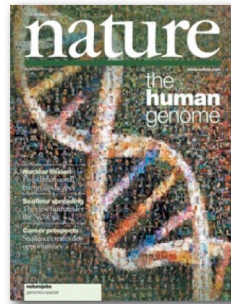
(the algorithms used in genomic analyses and applications)

- To understand what we can and cannot do with the data
- To generate data that is usable
- To know when to trust / not trust a genomic analysis
- To know what tools / methods / techniques are still missing

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# The Human Genome

Craig Venter    Francis Collins



READING THE BOOK OF LIFE: THE OVERVIEW

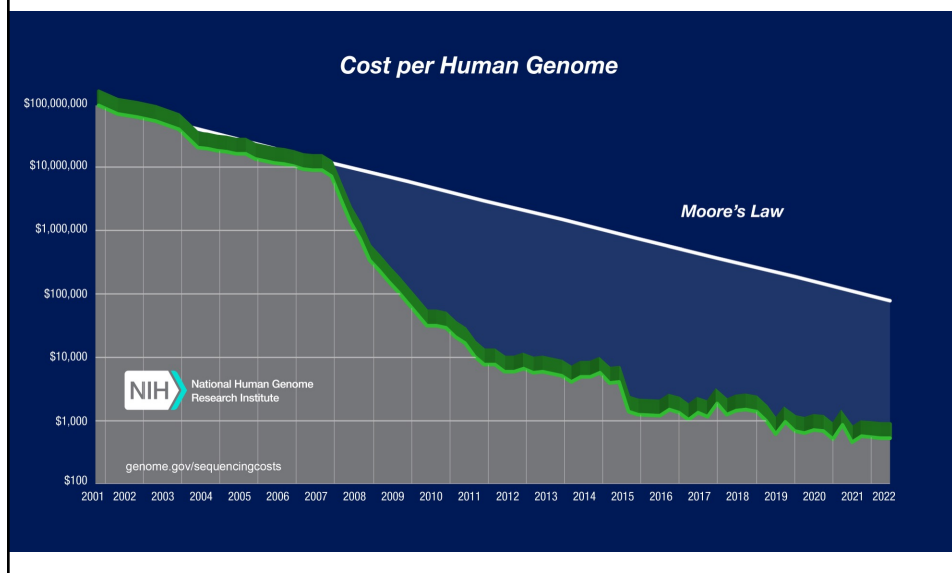
**READING THE BOOK OF LIFE: THE OVERVIEW;  
Genetic Code of Human Life Is Cracked by Scientists**

By NICHOLAS WADE  
Published: June 27, 2000

**The New York Times**

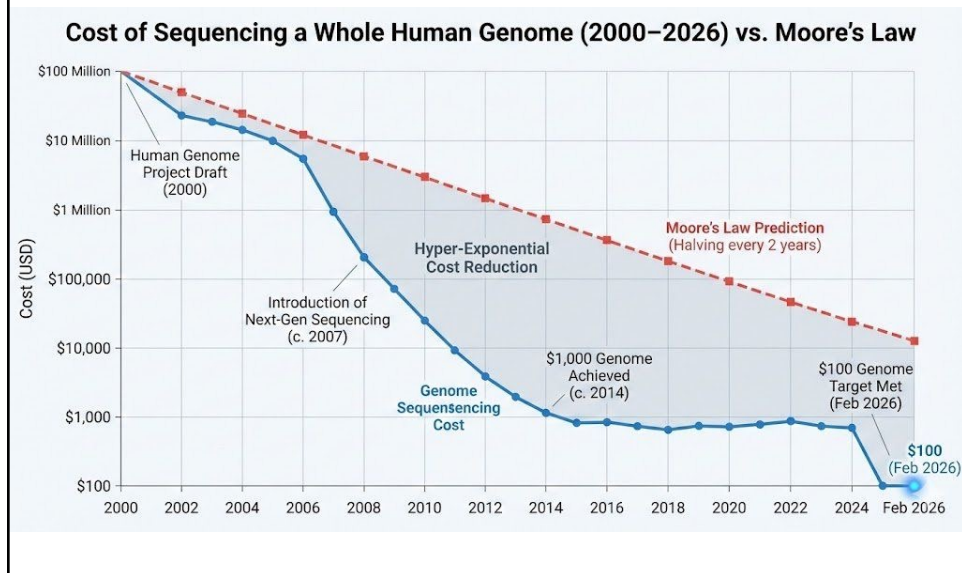
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## Sequencing Cost Per Human Genome



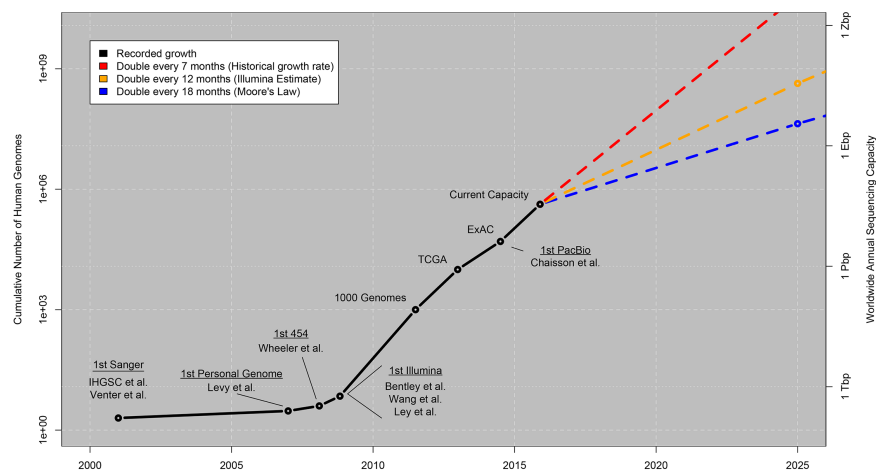
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## Sequencing Cost Per Human Genome



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## Sequencing Growth of Human Genomes



Stephens et al, Big Data: Astronomical or Genomical?, PLoS Biology, 2015

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1/1/2021

Eukaryotes(15469); Prokaryotes (304002); Viruses(42119)

3/05/2022

Eukaryotes (27961); Prokaryotes (511927); Viruses (56763)

2/27/2025

Eukaryotes (49030); Prokaryotes (2600000); Viruses (218200)

3/16/2026

Eukaryotes (60060); Prokaryotes (3000000); Viruses (258480)

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## How Big is Big Data?

### By 2025

| Data Phase          | Astronomy                                          | Twitter                     | YouTube                                              | Genomics                                                         |
|---------------------|----------------------------------------------------|-----------------------------|------------------------------------------------------|------------------------------------------------------------------|
| <b>Acquisition</b>  | 25 zetta-bytes/year                                | 0.5–15 billion tweets/year  | 500–900 million hours/year                           | 1 zetta-bases/year                                               |
| <b>Storage</b>      | 1 EB/year                                          | 1–17 PB/year                | 1–2 EB/year                                          | 2–40 EB/year                                                     |
| <b>Analysis</b>     | In situ data reduction                             | Topic and sentiment mining  | Limited requirements                                 | Heterogeneous data and analysis                                  |
|                     | Real-time processing                               | Metadata analysis           |                                                      | Variant calling, ~2 trillion central processing unit (CPU) hours |
|                     | Massive volumes                                    |                             |                                                      | All-pairs genome alignments, ~10,000 trillion CPU hours          |
| <b>Distribution</b> | Dedicated lines from antennae to server (600 TB/s) | Small units of distribution | Major component of modern user's bandwidth (10 MB/s) | Many small (10 MB/s) and fewer massive (10 TB/s) data movement   |

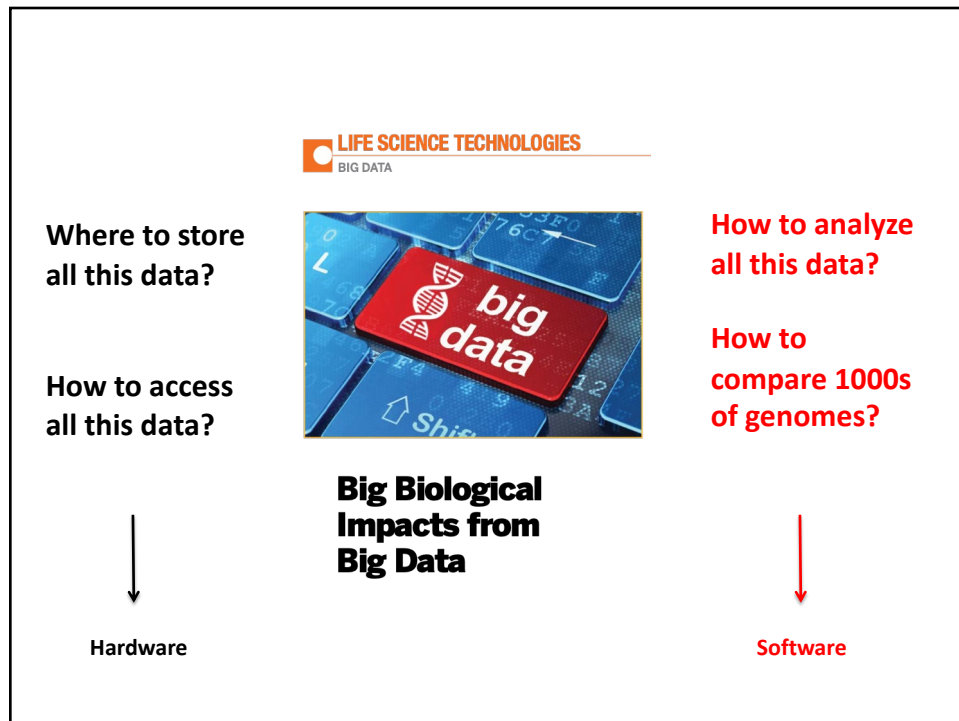
doi:10.1371/journal.pbio.1002195.t001

| Unit      | Size                          |
|-----------|-------------------------------|
| Byte      | 1                             |
| Kilobyte  | 1,000                         |
| Megabyte  | 1,000,000                     |
| Gigabyte  | 1,000,000,000                 |
| Terabyte  | 1,000,000,000,000             |
| Petabyte  | 1,000,000,000,000,000         |
| Exabyte   | 1,000,000,000,000,000,000     |
| Zettabyte | 1,000,000,000,000,000,000,000 |

Michael Schatz

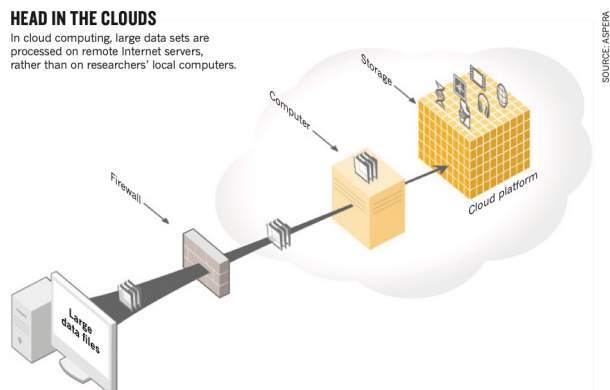
Stephens et al, Big Data: Astronomical or Genomical?, PLoS Biology, 2015

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## What is cloud computing?



Marx, The big challenges of Big data, Nature, 2013

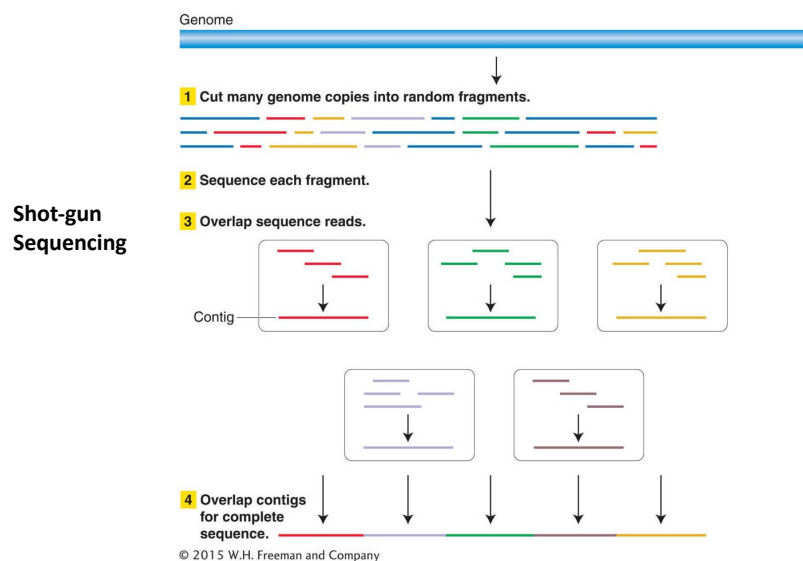
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## How to obtain a full genome?

- We need to “read” every single base in the genome, then put all the data back together into a completely ordered genome!
- Sequencing Technologies:
  - Sanger Sequencing
  - High-throughput sequencing, Next generation sequencing, Massively parallel sequencing:
    - Roche/454
    - Illumina Sequencing
    - Pacific Biosciences

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## The genome is very big!



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